

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 14

# What a Model Is and Why It Works

Parallel Reading

Self-Study Curriculum

**Parallel Reading:** PCM, section VII.2 (“Mathematical Biology”): read it not for the biology, which you know, but as a case study in model construction: at every step, notice what is being preserved and what is being discarded. Also read: George Box, “Science and Statistics” (1976, JASA), the paper where “all models are wrong” originates (the fuller “but some are useful” phrasing came in his 1979 follow-up). Read it, the paper is far more nuanced than the slogan.

Also read: Chapter 28 of MacKay, *Information Theory, Inference and Learning Algorithms* (“Model Comparison and Occam’s Razor”), approximately 12 pages.